

Olympiad Test Paper — Science (Class 7) — Paper 4

Chapter 18: Wastewater Story

Paper 4 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	18 — Wastewater Story
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Distractors exploit reversed aerobic/anaerobic roles, conflating cleaner with potable, swapping sludge/scum/grit, treating screening or settling as removing dissolved or microscopic load, mismatching by-products with their source process, and mistaking nutrient pollution and bacterial poisoning for harmless events. Do not write on this paper — mark answers on your answer sheet.

1. A field worker collects four liquids and must label which one is sewage. Which description matches sewage as defined in the chapter?

- (A) Rainwater that has fallen straight onto a clean roof and run into a barrel
- (B) A clear liquid from which every dissolved salt has already been removed
- (C) Used water carrying both impurities that float or settle and impurities dissolved invisibly within it
- (D) The dried solid cake left after dirty water has fully evaporated

2. A municipal map shows pipes carrying away water from kitchens, a dye-printing mill, irrigated fields, and street storm drains, all merging into one underground line. What does this single merger demonstrate?

- (A) Sewage is generated from domestic, industrial, agricultural and storm-water sources that share the collection network
- (B) Sewage can only ever originate inside household toilets
- (C) Sewage is manufactured at the treatment plant and pumped back to the city
- (D) Storm water is the only true source of sewage; the rest is clean

3. A sample of sewage is poured through filter paper, the filtrate is boiled until dry, and a white solid residue remains. The dissolved substance that produced this residue is best classified as which type of contaminant?

- (A) A floating organic scum that the filter paper trapped on top
- (B) A dissolved inorganic impurity such as a nitrate or phosphate salt
- (C) A living disease-causing microbe that survived the boiling
- (D) A heavy suspended grit particle that the filter held back

4. Of the materials below carried in household sewage, which one is the contaminant that is itself a living thing capable of multiplying and causing illness?

- (A) Dissolved phosphate from washing powder
- (B) Fine sand washed in from a courtyard
- (C) Floating grease from frying-pan rinse water
- (D) Disease-causing bacteria entering with human waste

5. A teacher lists four items found in sewage and asks the class to pick the one that is NOT organic matter. Which should they choose?

- (A) Leftover rice scraped from dinner plates
- (B) Faecal matter flushed from toilets
- (C) Mustard oil rinsed from a cooking vessel
- (D) Dissolved copper salt leaking from an electroplating workshop

6. Untreated sewage that is rich in dissolved nitrates and phosphates is discharged into a quiet pond. Following the chapter's reasoning about these 'nutrients', what is the most likely outcome?

- (A) The extra nutrients over-nourish water plants and upset the pond's natural balance
- (B) The nutrients make the pond water immediately fit for drinking
- (C) The nutrients sink and harden into harmless construction sand
- (D) The nutrients are converted by the pond into collectable biogas

7. A new colony is being planned. The engineer insists the sewer pipes must run UNDERGROUND and stay covered rather than as open channels along the road. The strongest health reason from the chapter is that covered sewers:

- (A) Allow more sunlight to disinfect the flowing sewage
- (B) Let aerobic bacteria in the open air clean the sewage faster
- (C) Keep disease-laden sewage from being exposed where it could contaminate water people use
- (D) Make the sewage produce biogas as it flows to the plant

8. Order the journey of a litre of dirty dishwater correctly: it is first received by the (i) sewer, treated at the (ii) treatment plant, and the cleaned outflow is sent to a (iii) river or used to recharge groundwater. Which arrangement of these stages is the actual order experienced?

(A) ii then i then iii

(B) iii then i then ii

(C) i then ii then iii

(D) ii then iii then i

9. At a plant the bar screens are the very first barrier. Which list contains ONLY things the bar screens are designed to stop?

(A) Dissolved nitrates, fine grit and disease bacteria

(B) Plastic bags, cloth rags and broken sticks

(C) Floating grease scum, settled sludge and biogas

(D) Fine sand, dissolved phosphate and cooking oil

10. A plant's bar screen has corroded away, so rags and plastic flow straight into the next units. Which problem is the MOST direct and earliest result?

(A) The pumps and channels just downstream become choked and damaged by the debris

(B) The dissolved nitrate level in the water suddenly doubles

(C) The treated outflow instantly becomes safe to drink

(D) Biogas output at the digester immediately triples

11. A student claims, 'After sewage clears the bar screen, only pure water is left.' What is the correct judgement?

(A) Wrong, because the screen removes only large objects while grit, finer solids, dissolved salts and microbes all stay behind

(B) Correct, because the bar screen filters out every impurity at once

(C) Correct, because all that sewage contains is large objects

(D) Wrong, because the bar screen actually adds new impurities to the water

12. In the grit-and-sand chamber the water is deliberately made to flow SLOWLY. What does the slowing accomplish here, and not elsewhere?

(A) It lets dissolved nitrate salts settle out of the water

(B) It allows floating oil to rise so it can be skimmed off

(C) It gives heavy sand, grit and pebbles enough time to sink to the floor of the chamber

(D) It encourages aerobic bacteria to multiply and feed

13. Why is the grit-and-sand removal placed BEFORE both the settling tank and the aeration tank in the sequence?

(A) Because grit is the most poisonous substance in all of sewage

(B) Because grit is the raw material that the digester turns into biogas

(C) Because the settling tank is physically unable to hold any solid at all

(D) So abrasive mineral particles do not wear out later equipment or get mixed into the organic sludge handled afterwards

14. During a cloudburst, sewers carry an unusually heavy load of washed-in road sand. Which single stage of the plant is built to cope with this and should be checked first?

- (A) The sludge digester (B) The aeration tank
(C) The grit-and-sand removal chamber (D) The bar screen

15. In the settling tank a long scraper crawls along the floor while a separate arm sweeps the surface. The floor scraper is moving which material?

- (A) The floating oily scum, lifting it off the top
(B) The settled sludge, pushing it toward the point where it is drawn off
(C) Bubbles of air being driven through to feed bacteria
(D) Large plastic bags caught at the tank's inlet

16. Two layers form in the settling tank: a dense layer at the bottom and a greasy film on top. Which naming of the two layers is correct?

- (A) Bottom layer is scum; top film is sludge (B) Bottom layer is biogas; top film is grit
(C) Both layers are together called manure (D) Bottom layer is sludge; top film is scum

17. An engineer needs to remove the fine suspended solids that slipped past the grit chamber. Which unit is designed to capture them?

- (A) The bar screen, whose bars sieve out the fine suspended solids
(B) The sludge digester, which strains the fine solids from the water
(C) The aeration tank, whose air current sweeps the fine solids to the floor
(D) The sedimentation (settling) tank, where the fine solids settle out as sludge

18. If the settling tank's bottom scraper jams and stops, what is the most direct consequence inside that tank?

- (A) Air stops bubbling through the aeration tank next door
(B) The bar screens at the inlet stop catching large debris
(C) Settled sludge piles up on the floor and is no longer carried away for digestion
(D) Biogas stops being released from the sealed digester dome

19. Air is forced through the aeration tank for a specific biological purpose. Which statement captures that purpose exactly?

- (A) It supplies oxygen so anaerobic bacteria can release biogas
(B) It supplies oxygen so aerobic bacteria can grow and consume the remaining dissolved organic matter
(C) It chills the water so that suspended grit settles faster
(D) It pushes large floating objects to where they can be screened

20. The settled mass of bacteria that has fed on the organic matter in the aeration tank is given a special name. What is it, and what is commonly done with most of it?

- (A) It is grit, and it is sent to make biogas in the digester
(B) It is activated sludge, and much of it is returned to the aeration tank to keep the bacterial action going

- (C) It is scum, and it is skimmed and burnt as fuel
- (D) It is raw sewage, and it is poured straight back into the inlet

21. The aeration tank's air blower fails and stays off for two days. Following the chapter's logic, what happens to the cleaning of the water?

- (A) Aerobic bacteria thrive even better without the disturbance of bubbling air
- (B) The tank begins producing collectable biogas in place of clean water
- (C) Gravity instantly filters the dissolved organic matter out by itself
- (D) Aerobic bacteria are starved of oxygen and break down the leftover organic waste poorly

22. Which observation alone would let a visitor confidently tell the aeration tank apart from the settling tank?

- (A) A sludge layer at the bottom and an oily scum on top, with no bubbling
- (B) Metal bars catching plastic bags at the entrance
- (C) Air is being vigorously bubbled through the water to grow bacteria, with no sludge-and-scum layering
- (D) A sealed dome from which flammable gas is being drawn off

23. The sludge from the settling tank is fed into a closed, air-tight digester rather than an open tank. What is the reason and result?

- (A) Aerobic bacteria can grow faster with all the trapped air inside
- (B) Anaerobic bacteria can act in the absence of air and release useful biogas
- (C) The sun can shine in and dry the sludge into manure on the spot
- (D) Large plastic items still in the sludge can finally be screened out

24. Match each by-product with the process that genuinely produces it.

- (A) Biogas comes from anaerobic digestion of sludge; manure comes from the dried digested sludge
- (B) Biogas comes from bubbling air through the aeration tank; manure is the floating scum
- (C) Biogas is given off by the bar screens; manure is the settled grit
- (D) Biogas is the same as the skimmed scum; manure is the activated sludge recycled to aeration

25. A village runs a community biogas unit fed with cattle dung and toilet waste. The bacteria generating its flammable gas work under which condition, and the same condition is found in which plant unit?

- (A) They work without air (anaerobic), the same as in the sealed sludge digester
- (B) They work with plenty of air (aerobic), the same as in the aeration tank
- (C) They work only in bright sunlight, the same as on the bar screens
- (D) They work only when frozen, the same as in the grit chamber

26. A student writes: 'Aerobic bacteria do the work inside the digester, and anaerobic bacteria do the work inside the aeration tank.' Why is this wrong?

- (A) Nothing is wrong; the description is exactly right
- (B) Only the digester half is wrong while the aeration half is right
- (C) Both tanks in fact use the very same aerobic bacteria
- (D) The roles are swapped; anaerobic bacteria act in the air-free digester and aerobic bacteria act in the aerated tank

27. The clear water leaving a plant is sometimes passed through a chlorination step before release. What is the point of this last step?

- (A) To destroy harmful microbes that the earlier physical and biological steps did not remove
- (B) To add nitrate nutrients so the released water is richer
- (C) To make the remaining heavy sand and grit settle out
- (D) To generate extra biogas from any leftover organic matter

28. Inlet (raw sewage) and outlet (treated) samples of a plant are compared. Which comparison is correct?

- (A) The outlet sample is dirtier than the inlet sample
- (B) The two samples are identical because treatment changes nothing
- (C) The outlet sample carries far less organic matter and far fewer solids than the inlet sample
- (D) The outlet sample is now guaranteed pure enough to drink straight away

29. A downstream village sees a plant releasing clear-looking treated water into the river and wants to drink it directly. Why is this unsafe?

- (A) The plant deliberately re-dirties the water just before releasing it
- (B) Treated water always contains more sewage than the inlet did
- (C) Treatment makes the water much cleaner but does not guarantee it is free of all germs and fit to drink
- (D) Rivers automatically purify any water that enters them

30. Arrange what LEAVES the main water stream from earliest to latest: (i) heavy grit, (ii) large floating debris, (iii) settled sludge with floating scum, (iv) remaining dissolved organic matter eaten by bacteria.

- (A) i then ii then iii then iv
- (B) iv then iii then i then ii
- (C) ii then i then iii then iv
- (D) iii then ii then i then iv

31. Sewage collected from a hospital, a tannery, a row of houses and a vegetable farm all reaches one plant. What does this tell us, and which contaminant type is the hospital most likely to add in excess?

- (A) Sewage comes only from homes, and a hospital adds only clean rainwater
- (B) Sewage is made at the plant, and a hospital adds only grit and sand

- (C) Sewage is just storm water, and a hospital adds only dissolved oxygen
- (D) Sewage has domestic, industrial and agricultural sources together, and a hospital adds extra disease-causing microbes

32. Trace the correct fate of the TWO streams that split inside the plant after the settling tank.

- (A) The settled sludge goes to the air-free digester, while the clarified water goes to the aerated tank
- (B) The settled sludge goes to aeration, while the clarified water goes to the air-free digester
- (C) Both the sludge and the water go together into the one sealed digester
- (D) Both the sludge and the water go together into the one aeration tank

33. A family pours a litre of hot used cooking oil down the kitchen sink every week. Which chain of effects best matches the chapter's warning?

- (A) The oil is the favourite food of aerobic bacteria and speeds the cleaning
- (B) The oil makes the treated water taste sweet and safe to drink
- (C) The oil instantly becomes useful manure while still inside the sewer
- (D) The oil coats and hardens inside the pipes, clogging them and making treatment harder

34. A factory quietly pipes strong chemicals straight into the public sewer. Beyond polluting the river, why is this especially damaging to the treatment PROCESS itself?

- (A) The chemicals make the bar screens trap debris more efficiently
- (B) The chemicals turn the sewage instantly into safe manure
- (C) The chemicals can kill the helpful bacteria that the plant depends on to clean the water
- (D) The chemicals increase the town's drinking-water supply

35. Among these household acts at the sink, which is LEAST likely to harm the treatment bacteria or block the pipes?

- (A) Tipping out leftover paint thinner (B) Pouring hot used frying oil
- (C) Flushing a handful of expired medicine tablets
- (D) Pouring plain water that was used to rinse a clean glass

36. A worker insists, 'Once the water clears the grit chamber, every solid has been taken out of it.' What is the correct judgement?

- (A) Right, because the grit chamber removes every solid completely
- (B) Wrong, because only the heavy sand and grit are gone while finer solids, scum and dissolved matter still remain
- (C) Right, because after grit removal only pure water is left
- (D) Wrong, because the grit chamber actually puts new solids into the water

37. Which statement correctly separates where DISSOLVED organic impurities are dealt with from where LARGE solids are dealt with?

- (A) Large solids are removed mechanically by screening and settling, while dissolved organic matter is consumed by bacteria during aeration
- (B) Both large solids and dissolved organic matter are caught by the bar screens
- (C) Dissolved impurities are caught in the grit chamber and the solids in the digester
- (D) Dissolved organic matter is skimmed off as scum and the solids are aerated

38. You watch flammable gas being safely drawn from a closed dome at one corner of a plant, with no air bubbling and no bars catching debris. Which unit are you observing?

- (A) The sealed sludge digester producing biogas
- (B) The aeration tank producing oxygen for its bacteria
- (C) The bar screen releasing gas from trapped debris
- (D) The settling tank giving off gas from its scum

39. Which step in the whole treatment removes impurities by a purely physical-mechanical action with NO help from bacteria?

- (A) The aeration tank consuming dissolved organic matter
- (B) The sludge digester decomposing the settled sludge
- (C) The bar screen catching large floating objects
- (D) The recycled activated sludge cleaning the water

40. A claim reads: 'The biogas from the digester and the scum from the settling tank are the same substance.' Judge it.

- (A) Wrong, because biogas is a flammable gas from anaerobic digestion while scum is a floating oily layer skimmed earlier
- (B) Right, because both are the floating oily film on the water
- (C) Right, because both are flammable gases drawn from sealed tanks
- (D) Wrong, because biogas is the bottom sludge and scum is a gas

41. Why does the chapter argue that even an excellent treatment plant is not enough on its own and home habits also matter?

- (A) Because the plant cleans water better when homes pour more oil into the drains
- (B) Because dumping oils, chemicals and solids down drains can clog the pipes and poison the bacteria the plant needs
- (C) Because home habits can rearrange the order of the treatment steps
- (D) Because households are expected to disinfect the river water themselves

42. A plant operator must clean treated effluent that is much clearer than the inlet, yet is told it still cannot be put into the town's drinking supply. The reason is that:

- (A) The operator deliberately keeps the effluent dirtier than the inlet for safety
- (B) Treatment greatly cuts solids and organic load but leaves no guarantee the water is germ-free and potable

- (C) Treated effluent is identical to bottled mineral water once it leaves the outlet
- (D) The drinking supply is required by law to be drawn only from sewage outlets

43. A town with poor toilets and open drains suffers repeated outbreaks of cholera and typhoid. The chapter points to which root cause, and which body's guidance frames sanitation worldwide?

- (A) Too much biogas is being made in homes, against the rules of the local bank
- (B) Sewage is mixing with the water people drink, against the guidance of the World Health Organisation
- (C) Treated water is being recharged into the ground, against the advice of the post office
- (D) The bar screens are catching too many plastics, against the orders of the fire brigade

44. A remote hill village has no underground sewer and wants each household to handle its own toilet waste cheaply on site. Which option fits, and what must eventually be done as solids build up inside it?

- (A) A full municipal treatment plant for each single house, which never needs emptying
- (B) A bar screen fitted under each toilet, which converts the sludge to drinking water
- (C) A septic tank, whose accumulated sludge must be periodically pumped out and disposed of or treated
- (D) A grit chamber for each house, whose sludge turns automatically into kitchen fuel

45. A vermi-processing toilet is promoted as environment-friendly. What makes it so, in contrast to a plain flush toilet emptying to a river?

- (A) It uses earthworms to turn the human waste into useful manure instead of releasing it untreated
- (B) It flushes the waste faster and further out into the nearest river
- (C) It burns the waste with no organisms to make biogas for the home
- (D) It pumps a constant stream of air to grow aerobic bacteria like a mini plant

46. At a remote campsite a chemical toilet is used instead of a sewer connection. Why are chemicals added to the waste there?

- (A) To turn the waste straight into safe drinking water on the spot
- (B) To grow aerobic bacteria that need a constant powered air pump
- (C) To collect flammable biogas piped to the camp kitchen
- (D) To break down the waste and cut odour and germs where no sewer or plant exists

47. A municipal crew regularly sweeps streets and cleans public drains and toilets. By the chapter's reasoning, how does this routine work protect public health?

- (A) It deliberately raises the number of microbes reaching household taps
- (B) It stops drains from overflowing and keeps disease-spreading waste away from where people live and gather

- (C) It adds extra chemicals into the public's household sinks
- (D) It speeds up the burning of biogas in people's kitchens

48. Both grit and sludge settle to a tank floor during treatment, yet they are handled in different tanks. Which contrast between them is correct?

- (A) Grit and sludge are the same material removed together in one tank
- (B) Grit is the floating oily layer and sludge is the flammable gas
- (C) Grit is the organic waste and sludge is the clean drinking water
- (D) Grit is heavy mineral sand removed early in its own chamber, while sludge is mostly organic solids settling later in the sedimentation tank

49. An engineer marks one stage as 'biological-anaerobic' and another as 'biological-aerobic'. Which pairing of each stage with its air condition and its product is fully correct?

- (A) The sludge digester is anaerobic and yields biogas; the aeration tank is aerobic and yields cleaned water with activated sludge
- (B) The sludge digester is aerobic and yields biogas; the aeration tank is anaerobic and yields manure
- (C) Both stages are anaerobic and both yield biogas
- (D) Both stages are aerobic and both yield activated sludge only

50. Treated effluent that is not fit for drinking is often pumped into the ground rather than discarded. Why is this a sensible use?

- (A) Because the effluent is already pure drinking water needing no care
- (B) Because the effluent is dirtier than raw sewage and must be hidden underground
- (C) Because soaking it underground converts the effluent into biogas
- (D) The cleaned water, though not potable, can safely soak into the ground and help replenish groundwater supplies

51. Place these four EVENTS a drop of sewage undergoes in the order they happen: (i) oily film skimmed as scum, (ii) plastic bag stopped, (iii) heavy sand sinks out, (iv) remaining organic matter eaten by aerobic bacteria.

- (A) i then ii then iii then iv
- (B) iv then iii then ii then i
- (C) iii then ii then i then iv
- (D) ii then iii then i then iv

52. Why does pouring solid food scraps and tea leaves down the sink cause trouble even though such matter is natural and organic?

- (A) They dissolve at once and improve the quality of the water
- (B) They turn straight into clean drinking water inside the pipe
- (C) They are eaten by aerobic bacteria that live in the home taps
- (D) Being solid, they build up and choke the pipes, obstructing the flow of sewage

53. Why does the chapter stress preventing sewage from contaminating water in the first place, rather than relying only on doctors to cure the resulting illnesses?

- (A) Because keeping sewage out of drinking water stops waterborne disease before it starts, which is better than only treating the sick
- (B) Because doctors are unable to treat any waterborne disease at all
- (C) Because clean drains generate more biogas for the hospitals
- (D) Because blocked drains somehow make drinking water safer

54. A plant must decide which products of the whole process are genuinely worth recovering and selling. Which pair is truly useful?

- (A) Plastic bags from the screen and grit from the chamber
- (B) Floating scum as a cooking fuel and grit as a fertiliser
- (C) Biogas as a fuel and dried digested sludge as manure
- (D) Raw untreated sludge tipped into the river as fish food

55. Why is it important that the grit-and-sand chamber acts on the water BEFORE it ever reaches the aeration tank?

- (A) Because the aerobic bacteria need the sand as their main food
- (B) So abrasive sand does not reach and wear out the aeration equipment or smother the working bacteria
- (C) Because the aeration tank removes grit better than the grit chamber does
- (D) Because grit has to be aerated before it is able to settle

56. Which description correctly identifies 'activated sludge'?

- (A) The large plastic debris caught on the bar screens
- (B) The settled mass of aerobic bacteria from the aeration tank that have cleaned the organic matter from the water
- (C) The flammable gas given off by the sealed digester
- (D) The heavy mineral sand settled in the grit chamber

57. A worker compares the bottom scraper of the SETTLING tank with the bottom of the AERATION tank. Which statement correctly distinguishes the bottom solids of each?

- (A) The settling tank's bottom holds raw settled sludge, while the aeration tank's bottom holds activated bacterial sludge
- (B) Both tank bottoms hold the same flammable biogas
- (C) Both tank bottoms hold heavy mineral grit only
- (D) The settling tank's bottom holds biogas, the aeration tank's bottom holds scum

58. Open defecation in fields near a stream is dangerous mainly because, according to the chapter:

- (A) The exposed waste quickly turns into safe biogas in the open air
- (B) Rain can wash the waste and its germs into the stream and other water sources, spreading disease
- (C) It increases the village's supply of clean drinking water
- (D) It feeds helpful aerobic bacteria living inside the village taps

59. A simplified plant skips its aeration tank entirely, sending settled, clarified water straight to the river. Which problem follows most directly from leaving out aeration?

- (A) Large plastic bags now flow uncaught into the river
- (B) The heavy grit and sand are no longer removed from the water
- (C) Dissolved organic matter is left in the water because the aerobic bacteria never got to consume it
- (D) The biogas supply from the digester is cut off completely

60. A chemist sorts the impurities of one sewage sample into three boxes: 'organic', 'dissolved inorganic' and 'living pathogens'. Into which box does each of human faeces, dissolved nitrate, and intestinal bacteria correctly go?

- (A) Faeces into dissolved inorganic, nitrate into living pathogens, and intestinal bacteria into organic
- (B) Faeces into organic, nitrate into dissolved inorganic, and intestinal bacteria into living pathogens
- (C) Faeces into living pathogens, nitrate into organic, and intestinal bacteria into dissolved inorganic
- (D) All three into the single box of dissolved inorganic impurities

Solutions — Science (Class 7), Chapter 18:

Wastewater Story — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	A	21	D	31	D	41	B	51	D
2	A	12	C	22	C	32	A	42	B	52	D
3	B	13	D	23	B	33	D	43	B	53	A
4	D	14	C	24	A	34	C	44	C	54	C
5	D	15	B	25	A	35	D	45	A	55	B
6	A	16	D	26	D	36	B	46	D	56	B
7	C	17	D	27	A	37	A	47	B	57	A
8	C	18	C	28	C	38	A	48	D	58	B
9	B	19	B	29	C	39	C	49	A	59	C
10	A	20	B	30	C	40	A	50	D	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

- (C)** Sewage is liquid waste water that carries suspended and dissolved impurities together; rainwater off a clean roof is not waste water, a salt-free clear liquid is not sewage, and a dried solid is the opposite of a liquid waste.
- (A)** The chapter lists homes, industry, farms and storm water as combined sources feeding the sewer, so the merger shows multiple sources; restricting it to toilets, claiming the plant makes it, or naming only storm water all contradict that list.
- (B)** Substances that pass the filter yet leave a solid on drying are dissolved inorganic salts like nitrates and phosphates; scum and grit would be caught by the filter, and a microbe would be destroyed by boiling and is not a dried salt residue.
- (D)** Among sewage contaminants only the pathogenic bacteria are alive and able to reproduce and infect; dissolved phosphate, sand and grease are all non-living impurities.

- 5. (D)** Organic matter comes from once-living material such as food, faeces and cooking oil, while a dissolved metal salt is an inorganic industrial impurity and is therefore the odd one out.
- 6. (A)** Nitrates and phosphates act as plant nutrients, so excess of them spurs overgrowth and disturbs the water body; they do not purify the water, become sand, or turn into biogas in a pond.
- 7. (C)** Covering the sewer prevents contact between germ-carrying sewage and the water and surroundings people rely on, which is the sanitation-disease link; sunlight disinfection, open-air aeration and in-pipe biogas are not why sewers are covered.
- 8. (C)** Dirty water leaves the home into the sewer, is then carried to the plant for cleaning, and only the treated outflow goes to a river or recharges groundwater; any order placing the plant or river before the sewer reverses the real path.
- 9. (B)** Bar screens trap only large floating objects such as bags, rags and sticks; dissolved salts, microbes, fine grit, scum, sludge, gas and oil all slip through and are dealt with by later stages.
- 10. (A)** Without screening, large debris jams and wears out the immediate downstream pumps and pipes; it cannot raise dissolved nitrates, cannot purify water, and would hinder rather than boost biogas.
- 11. (A)** Bar screens catch only big objects, so grit, dissolved matter and germs remain after them; the screen neither removes everything nor introduces impurities.
- 12. (C)** Reducing speed in the grit chamber lets dense mineral particles settle by gravity; dissolved salts do not settle there, oil-skimming belongs to the later settling tank, and bacterial feeding belongs to aeration.
- 13. (D)** Removing hard, abrasive grit early protects downstream machinery and keeps it out of the organic sludge stream; grit is not specially poisonous, is not feedstock for biogas, and the settling tank does hold settled sludge.
- 14. (C)** Sand and grit are captured in the dedicated grit chamber, so a sand surge stresses that unit; the digester, aeration tank and bar screen handle organic sludge, dissolved organics and large objects respectively.
- 15. (B)** The bottom scraper handles the heavy sludge that has settled, moving it for removal; skimming scum is the surface arm's job, while air-bubbling and bag-catching belong to the aeration tank and bar screen.
- 16. (D)** Heavy solids sinking to the floor are sludge while light grease floating up is scum; reversing them, naming gas or grit, or calling both manure are all incorrect.

- 17. (D)** Fine suspended solids that pass grit removal settle in the sedimentation tank as sludge; the bar screen only catches large objects, the digester acts on already-settled sludge, and aeration targets dissolved organic matter rather than sieving solids.
- 18. (C)** A jammed floor scraper leaves the settled sludge accumulating with no path to the digester; aeration air, bar-screen action and digester gas are governed by other equipment entirely.
- 19. (B)** Aeration feeds oxygen to aerobic bacteria that eat the leftover organic load; anaerobic bacteria avoid oxygen, cooling-for-settling is not the aim, and large-object removal happened at the bar screen.
- 20. (B)** The bacterial mass from aeration is activated sludge, a portion of which is recycled to the tank to sustain the cleaning bacteria; it is neither mineral grit, floating scum, nor raw incoming sewage.
- 21. (D)** Aerobic bacteria need the supplied oxygen, so cutting the air weakens them and leaves organic waste poorly treated; they do not benefit from losing oxygen, the tank is not a biogas unit, and dissolved organics do not settle by gravity.
- 22. (C)** Vigorous air bubbling is the signature of the aeration tank; quiet sludge-and-scum layering marks the settling tank, bar-catching marks the screen, and a gas-collecting dome marks the digester.
- 23. (B)** Sealing out air lets anaerobic bacteria decompose the sludge and give off biogas; aerobic bacteria need air, a sealed tank admits no sunlight, and plastics were already removed at the screen.
- 24. (A)** Anaerobic digestion of sludge yields biogas, and the leftover digested sludge dried out becomes manure; aeration makes neither gas, screens and grit chambers make no biogas, and scum is not biogas.
- 25. (A)** Biogas-making bacteria are anaerobic, mirroring the sealed sludge digester; aerobic conditions, sunlight-dependence and freezing are not how biogas bacteria operate.
- 26. (D)** The two are reversed: the digester is air-free and uses anaerobic bacteria while the aeration tank is supplied air for aerobic bacteria; the statement is not correct, not half-right, and the two tanks do not share one bacterial type.
- 27. (A)** Chlorination kills surviving disease microbes before discharge; it does not enrich the water with nutrients, settle grit, or make biogas.
- 28. (C)** Treatment strips out most organic matter and solids, so the outlet is much cleaner than the inlet; it is not dirtier, not unchanged, and still not certified drinkable.

29. (C) Treated effluent is cleaner yet not certified germ-free or potable, so it should not be drunk directly; the plant does not re-pollute the water, treated water is not dirtier than raw, and rivers do not self-purify reliably.

30. (C) Large debris goes first at the screen, then grit in its chamber, then sludge and scum in the settling tank, and finally dissolved organics are consumed in aeration; any order that puts grit before debris or aeration before settling is wrong.

31. (D) The mixed inputs show multiple sources, and a hospital's waste is rich in pathogens; sewage is not solely domestic, not made at the plant, and not mere storm water, and hospitals do not chiefly add clean rain, grit, or oxygen.

32. (A) Sludge is routed to the anaerobic digester and the clarified water to the aerobic aeration tank, treated separately; swapping their destinations or merging both streams into one tank misrepresents the process.

33. (D) Poured-away oil cools and hardens, choking pipes and burdening treatment; it does not boost bacterial cleaning, sweeten and purify water, or convert to manure in the pipe.

34. (C) Toxic chemicals poison the aerobic and anaerobic bacteria the plant relies on, crippling biological treatment; they do not improve screening, become manure, or add to drinking water.

35. (D) Plain rinse water adds nothing harmful, whereas paint thinner and medicines poison the bacteria and oil clogs the pipes.

36. (B) Grit removal takes out only heavy mineral particles, leaving finer suspended solids, scum and dissolved impurities for later units; it neither clears all solids nor adds any.

37. (A) Physical screening and sedimentation handle large solids, whereas dissolved organic matter is eaten biologically in aeration; screens cannot catch dissolved matter, the grit chamber does not capture dissolved impurities, and dissolved organics are not skimmable scum.

38. (A) A closed dome yielding flammable gas is the anaerobic sludge digester; the aeration tank bubbles air rather than collecting gas, and neither the bar screen nor the settling tank is a gas-producing sealed dome.

39. (C) Bar screening is a mechanical sieve needing no microbes, whereas aeration, digestion and activated-sludge action are all biological steps driven by bacteria.

40. (A) Biogas is a gas made by anaerobic digestion, while scum is a floating greasy solid removed in the settling tank, so they are different; they are not both oily films, not both

gases, and biogas is not bottom sludge.

41. (B) Careless drain dumping damages pipes and kills treatment bacteria, so good home habits protect the plant; pouring oil does not aid cleaning, home habits do not reorder plant steps, and households are not tasked with disinfecting rivers.

42. (B) Cleaner effluent is still not certified safe to drink, so it is barred from the drinking supply; it is not kept deliberately dirty, not equal to bottled water, and drinking supplies are not drawn from sewage outlets.

43. (B) Disease outbreaks trace to sewage contaminating drinking water, and the World Health Organisation is the body that frames sanitation; excess biogas, groundwater recharge and over-zealous screens are not the cause, and banks, post offices and fire brigades do not set sanitation guidance.

44. (C) A septic tank is the low-cost on-site choice and its settled sludge must be removed periodically; a full plant per house is impractical, a bar screen does not make drinking water, and a household grit chamber does not auto-produce fuel.

45. (A) A vermi-processing toilet relies on earthworms to convert waste into manure, keeping it out of rivers; it does not flush waste to a river, burn it for gas, or run an air pump for aerobic bacteria.

46. (D) Chemical toilets add chemicals to decompose waste and reduce smell and germs without any sewer; they do not produce drinking water, run an air pump for aerobes, or pipe biogas to a kitchen.

47. (B) Routine cleaning of drains and public toilets prevents overflow and contact with germ-laden waste, cutting disease; it does not raise tap microbes, dose home sinks with chemicals, or affect kitchen biogas.

48. (D) Grit is inorganic mineral sand caught early, whereas sludge is the organic solid that settles in the sedimentation tank; they are not one material, not an oily film and a gas, and not organic-waste-versus-drinking-water.

49. (A) The digester runs without air to make biogas while the aeration tank runs with air to clean water and grow activated sludge; the digester is not aerobic, the two stages are not both anaerobic, and they are not both aerobic with one product.

50. (D) Cleaned effluent is good enough to recharge groundwater even though it is not drinkable; it is not pure drinking water, not dirtier than raw sewage, and does not become biogas underground.

51. (D) The bag is caught first at the screen, then sand settles in the grit chamber, then scum is skimmed in the settling tank, and finally aerobic bacteria consume the leftover

organics; orders that skim scum before screening or place bacteria first are wrong.

52. (D) Solid scraps and leaves accumulate and block pipes regardless of being natural; they do not dissolve helpfully, become drinking water, or get eaten by bacteria inside home taps.

53. (A) Prevention by keeping sewage away from drinking water halts disease at the source, which beats curing it afterwards; doctors can treat such diseases, drains are not valued here for biogas, and blocked drains make water less safe.

54. (C) Biogas serves as fuel and the dried digested sludge as manure; bagged plastics and grit are waste, scum is not cooking fuel nor grit a fertiliser, and raw sludge dumped in a river is harmful, not useful.

55. (B) Removing grit before aeration protects the aeration machinery and bacteria from abrasive sand; bacteria do not feed on sand, the aeration tank is not a grit remover, and grit does not need aerating to settle.

56. (B) Activated sludge is the bacterial mass produced in aeration after consuming the organic load; it is not screen debris, digester gas, or grit-chamber sand.

57. (A) Raw sludge settles in the sedimentation tank while the aeration tank's floor collects activated (bacterial) sludge; neither bottom holds biogas or only grit, and neither holds scum at the floor.

58. (B) Rain carries the germ-laden waste into water sources, spreading waterborne disease, which is the danger of open defecation; the waste does not safely become biogas, add clean water, or feed tap bacteria.

59. (C) Aeration is where aerobic bacteria eat the dissolved organic load, so skipping it leaves that load in the water; large debris and grit are handled earlier by the screen and grit chamber, and biogas comes from the separate sludge digester.

60. (B) Faeces is once-living organic matter, nitrate is a dissolved inorganic salt, and intestinal bacteria are living pathogens, so each goes to its own box; the swapped pairings and lumping all three as inorganic ignore that the three are different kinds of contaminant.